

Backward Trajectory Completion in Bounded Phase Space: S-Entropy Framework, Virtual Sub-States, and the LHC Trigger as a Categorical Navigator

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Abstract

The LHC trigger faces a classification task: given a pattern of detector hits (the *endpoint*), determine the originating physics process (the *root*). We prove that this task is precisely backward trajectory completion in a ternary refinement hierarchy, and that it can be solved in $\mathcal{O}(\log_3 N)$ steps via the S-entropy embedding into $[0, 1]^3$ with the product Fisher metric — compared to $\Theta(N)$ for any algorithm restricted to physically realizable intermediate states.

The key mechanism is *virtual sub-states*: intermediate decompositions $(s_1, s_2, s_3) \in \mathbb{R}^3$ with mean $\bar{s} = s \in [0, 1]$ but at least one component outside $[0, 1]^3$. We prove three results about virtual sub-states that are central to trigger design: (i) they exist in positive Lebesgue measure (Theorem 6.2); (ii) restricting to physically realizable sub-states collapses backward completion to $\Theta(N)$ (Theorem 5.5); (iii) two backward trajectories sharing an endpoint are observationally indistinguishable from any metric invariant at the endpoint (Path Opacity, Theorem 7.1).

We establish the S-entropy framework from which these results derive: a bounded receiver $\mathcal{R} = (\Sigma, \Phi, K, \text{dec})$ has a strictly positive alignment floor $S_b(\mathcal{R}) > 0$ (Theorem 2.4), and three algebraic structures (oscillatory **Osc**, categorical **Cat**, partition **Part**) are mutually equivalent via explicit conversion functors (Theorem 3.5).

The paper concludes with the Aperture Identification theorem (Theorem 9.1): geometric aperture, virtual sub-state, miracle subtask, information catalyst, and Maxwell demon are five names for the same mathematical object — the partition-theoretic mechanism that enables $\mathcal{O}(\log N)$ navigation through the state space of physics events. This identifies the LHC trigger as a physical implementation of the categorical aperture, operating on collision events rather than molecular spectra.

Keywords: backward trajectory completion; S-entropy; virtual sub-states; ternary refinement hierarchy; LHC trigger; triple equivalence; path opacity; categorical aperture; information catalyst; partition algebra

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1 Introduction

1.1 The classification problem

The LHC trigger must classify 4×10^7 collision events per second. Each event is characterised by a pattern of detector hits: energies deposited in calorimeter cells, track momenta in the inner detector, muon hits in the outer spectrometer. The trigger must map this hit pattern to a class label (“interesting physics” or “background”) within $2.5 \mu\text{s}$.

Framed abstractly: the trigger is given the *endpoint* of a trajectory through the partition state space of the detector (the hit pattern) and must recover the *root* (the originating physics process). This is backward trajectory completion.

The conventional approach — forward simulation and threshold comparison — is computationally equivalent to forward trajectory construction: build all possible event topologies from assumed physics models and compare with the observed hit pattern. This is $\Theta(N)$ where N is the number of possible physics processes, which is astronomically large.

We show that backward trajectory completion, using the S-entropy embedding and virtual sub-states, achieves $\mathcal{O}(\log_3 N)$.

1.2 Outline and main results

Section 2 develops the S-entropy framework: bounded receivers, the S-functional, and the Floor Positivity theorem. Section 3 proves the Triple Equivalence between oscillatory, categorical, and partition descriptions. Section 4 defines the ternary refinement hierarchy and the S-entropy embedding. Section 5 proves backward navigation complexity and establishes the collapse theorem. Section 6 develops the theory of virtual sub-states. Section 7 proves path opacity and its trigger implications. Section 8 applies the theory to the LHC trigger: the detector as a backward navigator, the cell registry as an aperture, and the working example of the di-muon trigger. Section 9 proves the Aperture Identification theorem, unifying all five names for the same mechanism.

2 S-Entropy Framework and Receiver Theory

2.1 Bounded receivers

Definition 2.1 (Candidate and Truth Spaces). A *candidate space* $\mathcal{X}$ is a non-empty set of admissible solutions. A *truth space* $\mathcal{X}^* \subseteq \mathcal{X}$ is the designated set of correct

solutions. We allow $|\mathcal{X}^*| > 1$ to accommodate operationally indistinguishable solutions.

Definition 2.2 (Bounded Receiver). A *receiver* over $(\mathcal{X}, \mathcal{X}^*)$ is a quadruple $\mathcal{R} = (\Sigma, \Phi, K, \text{dec})$ where:

- (i) Σ is the *signal space*;
- (ii) $\Phi : \Sigma \rightarrow K$ is the *decoder*;
- (iii) K is the *knowledge framework* with operations $\{\circ_j\}$;
- (iv) $\text{dec} : K \rightarrow \mathcal{X}$ is the *candidate projection*.

The receiver is *bounded* if $|K| < |\mathcal{X}|$ (strictly smaller than the candidate space).

Definition 2.3 (S-Functional). An *S-functional* is a map $\mathcal{S} : \mathfrak{R} \times \mathcal{X} \times \mathcal{X}^* \rightarrow [0, 100]$ measuring misalignment: $\mathcal{S}_{\mathcal{R}}(x, x^*) = 0$ denotes perfect alignment; $\mathcal{S}_{\mathcal{R}}(x, x^*) = 100$ denotes maximum misalignment. It satisfies four axioms:

- (S1) Non-negativity: $\mathcal{S}_{\mathcal{R}}(x, x^*) \geq 0$.
- (S2) Boundedness: $\mathcal{S}_{\mathcal{R}}(x, x^*) \leq 100$.
- (S3) Self-truth lower bound: $\inf_x \mathcal{S}_{\mathcal{R}}(x, x^*) = S_b(\mathcal{R})$ for some function $S_b : \mathfrak{R} \rightarrow (0, 100]$.
- (S4) Receiver coherence: $\mathcal{S}_{\mathcal{R}}(x, x^*) = \delta_{\mathcal{R}}(\Phi(\sigma_x), \Phi(\sigma_{x^*}))$ for a non-decreasing $\delta_{\mathcal{R}}$ with $\delta_{\mathcal{R}}(\sigma, \sigma) = S_b(\mathcal{R})$.

2.2 The Floor Positivity theorem

Theorem 2.4 (Floor Positivity). *For every bounded receiver $\mathcal{R}$, $S_b(\mathcal{R}) > 0$.*

Proof. Suppose $S_b(\mathcal{R}) = 0$. By (S3), there exists $x_0 \in \mathcal{X}$ with $\mathcal{S}_{\mathcal{R}}(x_0, x^*) = 0$. By (S4), this requires $\delta_{\mathcal{R}}(\Phi(\sigma_{x_0}), \Phi(\sigma_{x^*})) = 0$. Since $\delta_{\mathcal{R}}(\sigma, \sigma) = S_b = 0$ and $\delta_{\mathcal{R}}$ is non-decreasing, $\Phi(\sigma_{x_0}) = \Phi(\sigma_{x^*})$ in K . The receiver is bounded: $|K| < |\mathcal{X}|$, so $|\text{dec}^{-1}(\{x^*\})| \leq |K| < |\mathcal{X}|$, leaving candidates unreachable in K . This contradicts the requirement that every candidate has a representative in K . Hence $S_b(\mathcal{R}) > 0$. $\square$

Corollary 2.5 (Smallest Knowable Error). *No candidate achieves $\mathcal{S}_{\mathcal{R}}(x, x^*) < S_b(\mathcal{R})$. The floor is intrinsic to the receiver: it depends on $|K|$, $\delta_{\mathcal{R}}$, and dec , not on the candidate or the truth.*

2.3 S-entropy coordinates

Definition 2.6 (S-Entropy Coordinate Space). The *S-entropy coordinate space* is the unit cube $\mathcal{S} = [0, 1]^3$ with coordinates (S_k, S_t, S_e) representing:

- S_k (*knowledge entropy*): normalised Shannon entropy of the frequency distribution of modes — how uniformly the knowledge is spread.
- S_t (*temporal entropy*): relative entropy of the phase position within the oscillation cycle.
- S_e (*evolution entropy*): entropy of the trajectory history — which sequence of states has been visited.

All three coordinates lie in $[0, 1]$ by normalisation; the cube $\mathcal{S}$ is a compact metric space under the Euclidean metric.

Theorem 2.7 (Ternary Encoding of S-Space). *Every point $\mathbf{S} \in \mathcal{S}$ corresponds to a unique infinite ternary string $\{t_i\}_{i \geq 1}$ with $t_i \in \{0, 1, 2\}$ encoding: $t_i = 0$ refines*

S_k ; $t_i = 1$ refines S_t ; $t_i = 2$ refines S_e . A k -trit string addresses one cell in the 3^k -fold partition of $\mathcal{S}$. As $k \rightarrow \infty$, the addressed cell converges to a unique point.

Proof. At each refinement step, the trit t_i selects which of three equal sub-intervals along the indicated axis contains the point. After k steps with n_j trits assigned to axis j ($j \in \{0, 1, 2\}$), the j -th coordinate is pinned to an interval of width 3^{-n_j} . Since $n_j \rightarrow \infty$ as $k \rightarrow \infty$ (each axis is refined at least $k/3$ times by the recurrence structure of bounded oscillations), all three interval widths $\rightarrow 0$ and the intersection is a single point. $\square$

3 Triple Equivalence

3.1 Three algebraic structures

Definition 3.1 (Oscillatory Structure). An *oscillatory structure* $\mathfrak{O} = (M, \omega, \Phi_{\mathfrak{O}}, \mu)$ consists of: mode set M ; frequency assignment $\omega : M \rightarrow \mathbb{R}_{\geq 0}$; phase function $\Phi_{\mathfrak{O}}(m, t) : M \times \mathbb{R} \rightarrow S^1$ with $\Phi_{\mathfrak{O}}(m, t + s) = \Phi_{\mathfrak{O}}(m, t) \cdot e^{2\pi i \omega(m)s}$; probability measure μ on M . Morphisms preserve frequencies and phase compatibility. The category is **Osc**.

Definition 3.2 (Categorical Structure). A *categorical structure* $\mathfrak{C} = (C, \prec, \lambda, \nu)$ consists of: cell set C ; strict partial order $\prec$ (refinement); labelling $\lambda : C \rightarrow \Lambda$; measure ν . Morphisms preserve $\prec$ and λ . The category is **Cat**.

Definition 3.3 (Partition Structure). A *partition structure* $\mathfrak{P} = (\Omega, \mathcal{P}, \pi)$ consists of: measurable space Ω ; refining sequence $\mathcal{P} = \{P_n\}_{n \in \mathbb{N}}$ of finite measurable partitions; measure π with $\pi(\partial p) = 0$ for each $p \in P_n$. Morphisms preserve refinement and measure. The category is **Part**.

3.2 Conversion functors

Definition 3.4 (Conversion Functors). Define:

- (i) $F_{OC} : \mathbf{Osc} \rightarrow \mathbf{Cat}$: sends $(M, \omega, \Phi_{\mathfrak{O}}, \mu)$ to $(M, \prec_{\omega}, \lambda_{\omega}, \mu)$ where $m_1 \prec_{\omega} m_2$ iff $\omega(m_1) | \omega(m_2)$ (rational ratio with bounded denominator), and $\lambda_{\omega}(m) = (\lfloor \log_2 \omega(m) \rfloor, \arg \Phi_{\mathfrak{O}}(m, 0))$.
- (ii) $F_{CP} : \mathbf{Cat} \rightarrow \mathbf{Part}$: sends $(C, \prec, \lambda, \nu)$ to $(C/\sim, \{P_n\}, \pi)$ where $\sim_n$ identifies cells agreeing on the first n label coordinates.
- (iii) $F_{PO} : \mathbf{Part} \rightarrow \mathbf{Osc}$: sends $(\Omega, \mathcal{P}, \pi)$ to $(P_{\infty}, \omega_{\mathfrak{P}}, \Phi_{\mathfrak{P}}, \pi)$ with $\omega_{\mathfrak{P}}(p) = 1/\pi(p)$ (Kac's lemma analogue).

Theorem 3.5 (Triple Equivalence). *The functors F_{OC} , F_{CP} , F_{PO} are part of mutually inverse equivalences of categories:*

$$F_{CO} \circ F_{OC} \cong \text{id}_{\mathbf{Osc}}, \quad F_{OC} \circ F_{CO} \cong \text{id}_{\mathbf{Cat}}, \quad (1)$$

$$F_{PC} \circ F_{CP} \cong \text{id}_{\mathbf{Cat}}, \quad F_{CP} \circ F_{PC} \cong \text{id}_{\mathbf{Part}}, \quad (2)$$

$$F_{OP} \circ F_{PO} \cong \text{id}_{\mathbf{Part}}, \quad F_{PO} \circ F_{OP} \cong \text{id}_{\mathbf{Osc}}. \quad (3)$$

Composing two of the three yields the third up to natural isomorphism.

Proof sketch. The inverse F_{CO} is constructed from F_{OC} by recovering the frequency from the first label coordinate ($\omega(c) = 2^{\lambda_1(c)}$) and the phase from the second ($\Phi(c, 0) = e^{i\lambda_2(c)}$). The composition $F_{CO} \circ F_{OC}$ acts on $(M, \omega, \Phi_{\mathfrak{O}}, \mu)$ and recovers the structure up to the discretisation error of $\lfloor \cdot \rfloor$, which vanishes as the labelling refines. Naturality is verified pointwise; the triangle identities follow from standard category theory (Mac Lane, 1998). The cyclic identities $F_{OC} \cong F_{PC} \circ F_{OP}$ etc. follow from uniqueness of equivalences up to natural isomorphism: any two functors inducing the same equivalence are naturally isomorphic. $\square$

Corollary 3.6 (Optimal Representation). *For any computation on a bounded oscillatory system, there exist equivalent formulations in all three representations (**Osc**, **Cat**, **Part**). The computation may be performed in whichever representation makes it computationally cheapest.*

Remark 3.7 (Physical meaning). Triple Equivalence is not merely algebraic. It says that measuring any one of: the oscillation frequency, the categorical state label, or the partition cell address — fully determines the other two. For the LHC trigger, this means: measuring the detector hit pattern (partition description) uniquely determines the oscillatory state (particle kinematics) and the categorical state (event class). The three representations are three windows onto the same physical reality.

4 Ternary Refinement Hierarchy and S-Entropy Embedding

4.1 The hierarchy

Definition 4.1 (Ternary Refinement Hierarchy). A *ternary refinement hierarchy* of depth k is a partition structure $\mathfrak{P} = (\Omega, \{P_j\}_{j=0}^k, \pi)$ where: $|P_0| = 1$ (the trivial partition), $|P_{j+1}| = 3|P_j|$ (each cell splits into exactly 3 children), and each block of P_j is a disjoint union of 3 blocks of P_{j+1} . The leaves are P_k with $|P_k| = 3^k = N$ elements.

Definition 4.2 (Trajectory in the Hierarchy). A *trajectory* in a depth- k hierarchy is a path $\mathcal{T} = (p_0, p_1, \dots, p_k)$ with $p_0 \in P_0$, p_{j+1} a child of p_j in P_{j+1} . The *endpoint* is $\partial(\mathcal{T}) = p_k \in P_k$. The *root* is p_0 .

Definition 4.3 (Completion Problem). The *trajectory completion problem* takes as input the hierarchy $\mathfrak{P}$ and an endpoint $p_k \in P_k$, and recovers the unique trajectory $\mathcal{T}$ with $\partial(\mathcal{T}) = p_k$.

4.2 The S-entropy embedding

Definition 4.4 (S-Entropy Embedding). The *S-entropy embedding* of a depth- k hierarchy is $\iota_k : P_k \rightarrow [0, 1]^3$ sending each leaf $p \in P_k$ to $\mathbf{S}(p) = (S_k(p), S_t(p), S_e(p))$

given by the base-3 expansion:

$$S_k(p) = \sum_{j=1}^k \frac{a_j^k(p)}{3^j}, \quad S_t(p) = \sum_{j=1}^k \frac{a_j^t(p)}{3^j}, \quad S_e(p) = \sum_{j=1}^k \frac{a_j^e(p)}{3^j}$$

where $a_j^c(p) \in \{0, 1, 2\}$ is the address digit of p at level j in the c -th coordinate.

Definition 4.5 (Product Fisher Metric). The *product Fisher metric* on $[0, 1]^3$ is the Riemannian metric

$$g(\mathbf{S}) = \sum_{c \in \{k, t, e\}} \frac{dS_c^2}{S_c(1 - S_c)}.$$

Theorem 4.6 (Embedding Necessity for Sub-linear Completion). *Any algorithm solving trajectory completion in $o(N)$ queries requires a metric on the leaf set P_k that distinguishes leaves at distance proportional to $\log_3 |P_k|$. The S -entropy embedding with the product Fisher metric provides exactly this.*

Proof. By the information-theoretic lower bound, any comparison-based algorithm on $N = 3^k$ leaves without geometric structure requires $\Omega(N)$ comparisons (each query reveals one bit of structure; there are $\Omega(N \log_2 3)$ bits to discover). With a metric whose ball of radius r contains at most $f(r)$ leaves, navigation requires $\mathcal{O}(\log f^{-1}(N))$ steps. For the S -entropy embedding with Fisher metric, the ball of radius r contains at most $\mathcal{O}(r^3 3^k)$ leaves (three-dimensional ball volume times leaf density), yielding $\mathcal{O}(\log_3 N)$ steps. $\square$

Theorem 4.7 (Recursive Self-Similarity). *The S -entropy embedding is strictly self-similar: for any block $b \in P_j$, the embedding restricted to leaves under b is isometric (after rescaling by $3^{-(k-j)}$) to the embedding of a depth- $(k-j)$ hierarchy.*

Proof. Leaves under b share the first j address digits; the remaining $k-j$ digits enumerate the sub-tree below b . Truncating the shared prefix and rescaling by $3^{-(k-j)}$ recovers the full depth- $(k-j)$ structure with the same Fisher metric. $\square$

5 Backward Navigation Complexity

5.1 Penultimate state and backward steps

Definition 5.1 (Penultimate State). For a trajectory $\mathcal{T} = (p_0, p_1, \dots, p_k)$ with $k \geq 1$, the *penultimate state* is p_{k-1} — the unique parent of p_k in P_{k-1} .

Theorem 5.2 (Penultimate Existence and Uniqueness). *For every leaf $p_k \in P_k$ with $k \geq 1$, the penultimate state p_{k-1} exists and is unique.*

Proof. Existence: the hierarchy is a rooted tree; every non-root node has a parent. Uniqueness: each block belongs to exactly one parent block (partitions are disjoint), so the parent is unique. $\square$

Theorem 5.3 (Backward Navigation Complexity). *The backward navigation algorithm from any leaf p_k to the root p_0 visits exactly $\log_3 N = k$ states and runs in $\mathcal{O}(\log_3 N)$ steps.*

Proof. Each backward step from p_j to p_{j-1} is implemented by dropping the last trit of the base-3 expansion of $\iota_j(p_j)$ — an $\mathcal{O}(1)$ operation. Starting at p_k and ending at p_0 requires exactly $k = \log_3 N$ steps. $\square$

The backward navigation algorithm is:

Algorithm 1 Backward trajectory completion

Require: Hierarchy $\mathfrak{P}$, leaf p_k

Ensure: Full trajectory $(p_0, p_1, \dots, p_k)$

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1:  $\mathcal{T} \leftarrow [p_k]$ 
2:  $j \leftarrow k$ 
3: while  $j > 0$  do
4:    $p_{j-1} \leftarrow \text{PARENT}(\mathfrak{P}, p_j)$   $\{\mathcal{O}(1)$  via base-3 truncation $\}$ 
5:    $\mathcal{T}.\text{PREPEND}(p_{j-1})$ 
6:    $j \leftarrow j - 1$ 
7: end while
8: return  $\mathcal{T}$ 
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5.2 Forward asymmetry

Theorem 5.4 (Forward Asymmetry). *Forward trajectory construction (root to leaf) cannot use virtual sub-states. Only backward completion (leaf to root) can.*

Proof. Forward construction at step j produces p_{j+1} as a refinement of p_j ; the system must commit to $p_{j+1} \in P_{j+1}$ as a physical block, hence $\iota_{j+1}(p_{j+1}) \in [0, 1]^3$. Virtual sub-coordinates $(s_1, s_2, s_3) \notin [0, 1]^3$ do not correspond to any partition block and cannot serve as p_{j+1} in a forward step.

Backward completion, by contrast, computes intermediate decompositions *analytically* within the receiver's internal calculus. The receiver's evaluation map permits any decomposition satisfying mean-recovery (Definition 6.1); the partition requires only that the global coordinate at each level lies in $[0, 1]^3$. The virtual sub-coordinates exist in the receiver's internal computation, not in the physical partition. $\square$

5.3 Collapse without virtual states

Theorem 5.5 (Collapse Without Virtual States). *Backward trajectory completion restricted entirely to physically realizable sub-coordinates (all intermediate states in $[0, 1]^3$) has worst-case complexity $\Theta(N)$, identical to naive forward enumeration.*

Proof. A physical-only backward step at depth j requires identifying p_{j-1} from p_j without the geometric shortcut provided by the Fisher metric. The algorithm must enumerate the children of candidate parent blocks to verify which contains p_j .

At level j , there are 3^j candidate parent blocks; each has 3 children. Without the embedded coordinates,

verifying parenthood requires $\mathcal{O}(3^j)$ comparisons per level. Summing over all levels: $\sum_{j=0}^k 3^j = (3^{k+1} - 1)/2 = \Theta(N)$.

More precisely: without geometric structure, the algorithm must construct the hierarchy from scratch by queries, which requires $\Omega(N \log_2 3)$ bits of information (Theorem 4.6) and cannot be done in $o(N)$ queries. $\square$

Corollary 5.6 (Virtual States are Necessary for Efficiency). *Virtual sub-states are not optional features of backward trajectory completion. Restricting to physical intermediate states increases the computational cost from $\mathcal{O}(\log N)$ to $\Theta(N)$ — for $N = T(60, 3) \approx 10^{59}$ LHC event categories, this is the difference between microseconds and cosmological timescales.*

6 Virtual Sub-States

6.1 Definition and existence

Definition 6.1 (Sub-Coordinate Decomposition). A sub-coordinate decomposition of a global coordinate $s \in [0, 1]$ is an ordered triple $(s_1, s_2, s_3) \in \mathbb{R}^3$ satisfying the mean-recovery constraint:

$$\bar{s} = \frac{s_1 + s_2 + s_3}{3} = s.$$

The decomposition is *physical* if $(s_1, s_2, s_3) \in [0, 1]^3$. It is *virtual* if at least one $s_i \notin [0, 1]$.

Theorem 6.2 (Virtual Sub-State Existence). *For every $s \in (0, 1)$ and every $\varepsilon > 0$, the set of virtual sub-coordinate decompositions with mean s and $\max_i |s_i| \leq M$ (for any $M \geq \varepsilon$) has positive Lebesgue measure in the constraint plane $\{(s_1, s_2, s_3) : (s_1 + s_2 + s_3)/3 = s\}$. Moreover, the measure of virtual decompositions approaches 1 as $M \rightarrow \infty$.*

Proof. The constraint plane is two-dimensional in $\mathbb{R}^3$. The physical region (intersection with $[0, 1]^3$) is a convex polygon of bounded area. The virtual region (outside $[0, 1]^3$) grows as M^2 with M . The fraction of the constraint plane that is physical tends to 0 as $M \rightarrow \infty$, so the virtual fraction tends to 1. For any fixed $M \geq \varepsilon$, the specific set $\{(s + \varepsilon, s + \varepsilon, s - 2\varepsilon) : \varepsilon \in (0, M - s)\}$ contains virtual decompositions (first two components exceed 1 for $\varepsilon > 1 - s$) of positive measure. $\square$

Theorem 6.3 (Local-Global Decoupling for Trajectories). *Let $\mathcal{T} = (p_0, \dots, p_k)$ be a trajectory with endpoint $\iota_k(p_k) = (S_k, S_t, S_e) \in [0, 1]^3$. For any target sequence of sub-coordinate decompositions $((s_1^{(j)}, s_2^{(j)}, s_3^{(j)}))_{j=0}^k$ with mean-recovery $\bar{s}^{(j)} = \iota_j(p_j)_c$ (one component c per level), there exists a receiver-internal computation realising the trajectory with these sub-coordinates at each step.*

Proof. Apply the Unconstrained Subtask theorem (Sachikonye, d) per step: the global S-value at step j is the embedded coordinate; the local sub-coordinate decomposition is a free choice subject to mean-recovery;

the cancellation pattern $(s_i - s_i)$ preserves the global value. Construct level by level. $\square$

Corollary 6.4 (Miracle Trajectory). *For any trajectory $\mathcal{T}$, there exists a receiver-internal computation realising $\mathcal{T}$ in which every intermediate sub-coordinate is virtual (outside $[0, 1]$).*

Proof. At each step, choose sub-coordinates $(M + \varepsilon, M + \varepsilon, 3s - 2M - 2\varepsilon)$ with M large. The first two components exceed 1; the third is highly negative. The mean is s . $\square$

Remark 6.5 (Physical interpretation). The Miracle Corollary has a striking physical interpretation. A trigger that classifies an event as a $Z \rightarrow \mu^+ \mu^-$ decay can, internally, pass through sub-computations that are “completely wrong” (S-value = 100 for each individual step) while ultimately arriving at the correct classification. This is not a failure of the algorithm — it is the mechanism that makes efficient classification possible. The virtual sub-states are not errors; they are the essential navigational tools.

6.2 The miracle principle

Theorem 6.6 (Miracle Principle). *For any target global S-value $\mathcal{S}^* \in [S_b(\mathcal{R}), 100]$ and any assignment of local S-values $(\sigma_1, \dots, \sigma_n) \in [S_b, 100]^n$, there exists a computation realising $\mathcal{S}^*$ globally with $\mathcal{S}(\eta_i) = \sigma_i$ locally for each subtask η_i .*

Proof. Construct the global computation as $\xi = \xi_0 + \sum_i (\eta_i - \eta_i)$ where ξ_0 has global S-value $\mathcal{S}^*$ and each η_i is chosen with local value σ_i . The cancellations $\eta_i - \eta_i = 0$ contribute nothing to the evaluation; the local S-values are by construction the chosen targets. $\square$

Corollary 6.7 (Miraculous Subtasks). *Taking $\sigma_i = 100$ for all i gives a globally correct computation in which every subtask is maximally wrong locally. These are miraculous subtasks: individually wrong, collectively right.*

7 Path Opacity and Its Trigger Implications

Theorem 7.1 (Path Opacity). *Let $\mathcal{T}_1$ and $\mathcal{T}_2$ be two backward trajectories sharing the same endpoint $p_k \in P_k$ but with distinct sequences of virtual sub-coordinate decompositions at intermediate steps. Given only the endpoint p_k and the metric g , no metric invariant computed at p_k distinguishes $\mathcal{T}_1$ from $\mathcal{T}_2$.*

Proof. By Theorem 6.3, intermediate sub-coordinate decompositions are unconstrained subject to mean-recovery; any two trajectories with the same endpoint but distinct intermediates both satisfy the trajectory definition.

Every metric invariant $I(\mathcal{T}, g)$ that depends on the endpoint alone — e.g., $\mathbf{S}(p_k)$, the geodesic distance $d(p_0, p_k)$, the Riemannian volume of the constraint plane through p_k — is by definition equal for both trajectories.

Invariants depending on intermediate states are not accessible from the endpoint; they cannot be computed from p_k without additional information about the path taken. $\square$

Corollary 7.2 (S-matrix Analog). *Path opacity is the categorical analog of S-matrix theory: only the initial and final states (root and endpoint) are observable. The virtual intermediate states are unobservable. This is not an accident: it is the structural reason that quantum field theory’s S-matrix can be computed from Feynman diagrams that involve unphysical (off-shell, virtual) intermediate states without contradiction.*

Remark 7.3 (Trigger implication). Path opacity means that the trigger’s internal computation — which virtual sub-state decompositions it uses, which intermediate evaluations it performs — is invisible to any external observer and irrelevant to the classification result. The trigger may use any internally convenient virtual computations as long as it arrives at the correct endpoint classification. This is the formal justification for why the HLT can use approximations, look-up tables, and cached partial results: the internal path does not affect the final classification.

Theorem 7.4 (Non-Demolition Classification). *The backward completion algorithm makes exactly one observable: the root category p_0 . The intermediate virtual sub-states are computed receiver-internally and never become physical observables. The classification is therefore non-demolition: it extracts category information from the endpoint without creating any observable disturbance to the intermediate computation.*

Proof. The backward algorithm (Section 5) reads the trit string of $\iota_k(p_k)$ and outputs p_0 (the root category). The virtual sub-coordinates, which exist only in the receiver’s internal S-space computation, do not appear in any physical output. The classification maps $p_k \mapsto p_0$ deterministically and completely; no disturbance to p_k or to the physical state at the endpoint is required or produced. $\square$

8 The LHC Trigger as a Categorical Navigator

8.1 Event as endpoint, physics as root

The LHC trigger classification problem maps exactly onto the trajectory completion problem:

Trajectory completion	LHC trigger
Leaf p_k (endpoint)	Detector hit pattern
Root p_0	Originating physics process
Hierarchy depth k	Number of detector subsystems (slightly higher: includes the post-decay kinematics).
Backward navigation	Trigger classification
Virtual sub-states	Internal trigger computations
Path opacity	HLT internal computations invisible to the external world.
$\mathcal{O}(\log_3 N)$	Real-time classification in $< 2.5 \mu\text{s}$ (collision).

Definition 8.1 (LHC Event Hierarchy). The LHC event classification hierarchy is a ternary refinement hierarchy of depth $k = 3$ (current ATLAS) to $k = 4$ (HL-LHC) with levels:

- (1) Level 0 (root): broad physics category (QCD background, electroweak, Higgs).
- (2) Level 1: specific process type (dijet, W/Z production, $t\bar{t}$, Higgs).
- (3) Level 2: decay channel ($W \rightarrow \ell\nu$, $Z \rightarrow \ell\ell$, $H \rightarrow \gamma\gamma$).
- (4) Level 3 (leaf): specific detector hit pattern (trigger condition fired).

Each level trifurcates: three broad categories at level 0, nine at level 1, etc.

8.2 Cell registry as the S-entropy embedding

Theorem 8.2 (Physical Origin of the Trigger Cell Registry). *The trigger’s cell registry $\Gamma = \{C_i\}$ is the physical realisation of the S-entropy embedding ι_k : each cell C_i corresponds to a cell in the 3^k -fold partition of $[0, 1]^3$, and the partition coordinates (n, ℓ, m, s) of the detected particle determine which cell C_i it falls into.*

Proof. From Sachikonye (b) and Sachikonye (c), the partition coordinates (n, ℓ, m, s) label the cells of the ternary refinement hierarchy at depth $k = 4$ (for the four subsystems tracker/ n , ECAL/ ℓ , HCAL/ m , muon/ s). The S-entropy embedding maps each combination of partition coordinates to a unique point in $[0, 1]^3$ via the base-3 expansion of Definition 4.4.

The trigger’s timing cell $C_i = \{|\Delta P| < \delta_i\}$ constrains the S_t coordinate of the event (temporal entropy — how close the signals are to the reference bunch crossing). Combined with the ECAL shower shape (constraining S_e) and track momentum (constraining S_k), the three coordinates together uniquely address a leaf in the S-entropy cube. $\square$

8.3 The di-muon trigger as a worked example

The di-muon trigger ($Z \rightarrow \mu^+\mu^-$ selection) is the simplest physics-motivated trigger and the natural starting point for illustration.

Example 8.3 (Di-Muon Backward Completion). A collision event produces two muons of opposite sign with $p_T > 20 \text{ GeV}$. The backward completion proceeds:

Level 3 (endpoint): The muon system fires for two tracks in opposite hemispheres. In the S-entropy embedding: $S_k \approx 0.7$ (high momentum tracks, near-maximum energy shell), $S_t \approx 0.5$ (signals arrive on-time, at the median temporal entropy), $S_e \approx 0.3$ (low evolution entropy, clean two-track topology).

Level 2 (backward step 1): Parent cell has $S_e \approx 0.33$ (slightly higher: includes the post-decay kinematics). This corresponds to the $Z \rightarrow \ell^+\ell^-$ decay channel.

Level 1 (backward step 2): Parent cell has $S_e \approx 0.5$ (invisible intermediate evolution: the Z boson produced in the collision). This corresponds to Drell-Yan Z production.

Level 0 (root): $S_e \approx 0.8$ (high evolution entropy: the original quark-antiquark annihilation). This is the electroweak production category.

The three backward steps ($k = 3$) recover the complete physics process from the detector hit pattern in exactly 3 operations.

The virtual sub-states are the intermediate decompositions: at Level 2 $\rightarrow$ Level 1, the computation internally uses $(s_1, s_2, s_3) = (0.6, 0.4, -0.1)$ for the S_e coordinate (the -0.1 is virtual). The mean $(-0.1 + 0.6 + 0.4)/3 \approx 0.3$ recovers the physical value. The virtual component is invisible to any external measurement.

8.4 Why selection rules are structural dead zones

From the partition coordinate structure (Sachikonye, b), selection rules $\Delta\ell = \pm 1$, $\Delta m \in \{0, \pm 1\}$ constrain which transitions between partition states are physically realizable. In the backward completion framework, this has a direct consequence:

Theorem 8.4 (Selection Rules as Dead Zones). *Trigger backgrounds that require selection-rule-violating intermediate transitions between partition states cannot be reached by backward completion from any physically realizable endpoint.*

Proof. In the ternary hierarchy, a transition $\Delta\ell = \pm 2$ (for example) would require a parent cell at level $j - 1$ that is two angular levels removed from the child cell at level j . By the partition capacity formula $C(n) = 2n^2$ and the $SO(3)$ structure of the partition coordinates (Sachikonye, b), no two adjacent levels in the refinement hierarchy are connected by $|\Delta\ell| \geq 2$ transitions. Therefore, no backward step from a physical leaf p_k can traverse such a transition: it would require a non-existent parent block.

Formally: the backward algorithm uses only the parent map $p_j \mapsto p_{j-1}$, which by definition respects the refinement order. The refinement order imposes exactly the selection rules. Backgrounds requiring forbidden transitions are not merely rare — they have no pre-image under the parent map and thus have zero probability under any backward completion from a physical endpoint. $\square$

Remark 8.5 (Type error vs. probabilistic suppression). The difference between Theorem 8.4 and conventional background suppression is categorical: conventional triggers suppress backgrounds by imposing energy thresholds (probabilistic); the partition framework excludes backgrounds by structural non-existence (categorical). A background requiring $\Delta\ell = 2$ is not “rare” — it is absent from the backward completion tree entirely.

9 The Aperture Identification Theorem

9.1 Five names for one object

The following five descriptions appear in different branches of physics, mathematics, and computer science. We prove they refer to the same mathematical object.

Theorem 9.1 (Aperture Identification). *The following five descriptions name the same mathematical object, which we call the categorical aperture:*

- (1) **Geometric aperture:** the convex hull of admissible sub-coordinate decompositions of a fixed global coordinate, intersected with the virtual region $\mathbb{R}^3 \setminus [0, 1]^3$.
- (2) **Virtual sub-state:** a non-physical intermediate decomposition $(s_1, s_2, s_3) \notin [0, 1]^3$ with mean $s \in [0, 1]$, used in backward trajectory completion (Section 6).
- (3) **Miracle subtask:** a subtask η in a computation with local S -value $\mathcal{S}(\eta) = 100$ (maximally misaligned) but contributing to a globally aligned composition (Theorem 6.6).
- (4) **Information catalyst:** an expression atom whose application strictly decreases the receiver’s S -distance to truth, $\mathcal{S}(\xi \diamond \gamma, x^*) < \mathcal{S}(\xi, x^*)$ (Sachikonye, d).
- (5) **Corrected Maxwell demon:** a sorting element that selects which partition cells are accessible based on their categorical addresses, operating without thermodynamic work (from Axiom ??, the categorical observable commutes with the physical Hamiltonian).

All five share four defining properties: enable a transition; not consumed in enabling it; transfer zero information about themselves to the transition outcome; are necessary for the $\mathcal{O}(\log N)$ speedup.

Proof. We exhibit explicit identifications between each pair.

(1) $\leftrightarrow$ (2): The convex hull of admissible decompositions of a global coordinate s intersected with $\mathbb{R}^3 \setminus [0, 1]^3$ is precisely the set of virtual sub-states with mean s . This is immediate from Definition 6.1.

(2) $\leftrightarrow$ (3): A virtual decomposition $(s_1, s_2, s_3) \notin [0, 1]^3$ has at least one component s_i that, as an independent sub-coordinate, would have $\mathcal{S}(s_i, x^*) = 100$ (it lies outside the physical state space, maximally misaligned). This is the miracle subtask: individually $\mathcal{S} = 100$, collectively the mean recovers the correct value.

(3) $\leftrightarrow$ (4): A catalyst γ is an expression whose application reduces the global S -value. The Miracle Principle (Theorem 6.6) gives catalyst-like constructions; conversely, any catalyst can be realised as a virtual sub-state decomposition (Theorem 6.3).

(4) $\leftrightarrow$ (5): An information catalyst that operates on categorical addresses rather than physical quantities is precisely a sorting element that uses geometric (positional) information about categorical cells to direct transitions. By the non-demolition theorem (Theo-

rem 7.4), this sorting requires zero thermodynamic work (it operates on the categorical degree of freedom, not the physical degree of freedom). This is the corrected Maxwell demon: not a paradox, but a physical sorting element that exploits the commutativity $[\hat{O}_{\text{cat}}, \hat{O}_{\text{phys}}] = 0$ from Sachikonye (c).

Shared properties: All five enable transitions by providing geometric shortcuts through state space. None is consumed: the virtual sub-state exists as long as its mean-recovery constraint is satisfied; the catalyst can be applied multiple times. None transfers information about itself: path opacity (Theorem 7.1) shows the virtual path is invisible. All are necessary: Theorem 5.5 shows that without virtual states, complexity collapses to $\Theta(N)$. $\square$

Corollary 9.2 (The LHC Trigger is a Categorical Aperture). *The LHC trigger implements the categorical aperture. Its timing cells define which S-entropy regions are accessible (geometric aperture). Its internal HLT computations use virtual intermediate states (virtual sub-states). Its trigger conditions are individually approximate (miracle subtasks: each condition has non-zero false positive rate). The trigger collectively achieves precise event classification (globally aligned composition). It operates without disturbing the classified event (non-demolition). And it enables real-time classification that would be impossible by forward enumeration (necessary for the $\mathcal{O}(\log N)$ speedup).*

10 Discussion

10.1 Complexity summary

The main complexity results are:

- Backward completion with virtual sub-states: $\mathcal{O}(\log_3 N)$ (Theorem 5.3).
- Backward completion restricted to physical states: $\Theta(N)$ (Theorem 5.5).
- Forward enumeration (conventional trigger approach): $\Theta(N)$.

For $N = T(60, 3) \approx 10^{59}$ LHC event categories, the difference between $\log_3(10^{59}) = 98$ steps and 10^{59} steps is the difference between real-time trigger decisions and physically impossible computation.

10.2 The S-entropy floor as trigger precision floor

The Floor Positivity theorem (Theorem 2.4) has a direct trigger interpretation. No bounded trigger system can achieve $\mathcal{S}(\text{trigger}(x), x^*) = 0$ for all events: the trigger's finite knowledge framework K cannot represent every possible event topology. The floor $S_b(\mathcal{R}) > 0$ quantifies the irreducible misclassification probability of any trigger of given complexity.

For the ATLAS L1 trigger with $\sim 10^3$ trigger paths out of $\sim 10^{59}$ possible topologies, the composition efficiency $\eta_C \approx 0.08$ from Sachikonye (c) quantifies how far the trigger is from the theoretical floor. Increasing

η_C toward 1 (filling the $n_P = 60$ depth) would reduce the floor and decrease the irreducible misclassification probability.

10.3 Relation to Kalman filtering

The HLT track reconstruction uses the Kalman filter, which is a forward-backward algorithm: forward pass accumulates state estimates; backward pass (smoothing) corrects them using future measurements. The backward pass is precisely backward trajectory completion in the particle trajectory state space. The virtual intermediate states in the backward completion are exactly the extrapolated state estimates that lie outside the detector acceptance in intermediate layers — “virtual” in the sense that no detector exists at that position, yet essential for the smoothed estimate.

This observation suggests that Kalman filtering and S-entropy backward completion are the same algorithm in different coordinate systems, related by the Triple Equivalence (Theorem 3.5).

10.4 Open questions

Path multiplicity: For a fixed endpoint p_k , how many distinct backward trajectories exist in the ternary hierarchy? By the Path Opacity theorem, they are all equivalent from the observer's perspective, but their total count bounds the trigger's internal computational freedom.

Optimal virtual state selection: Among all virtual sub-state decompositions achieving mean-recovery, which minimise the Fisher information along the backward path? This would give an optimal backward navigation algorithm rather than the generic one of Algorithm 1.

Cascade catalysis and trigger iterations: The multiplicative catalyst algebra (proved in Sachikonye (a)) says catalyst power combines as $\kappa(\gamma_1 \diamond \gamma_2) = 1 - (1 - \kappa_1)(1 - \kappa_2)$. A trigger menu of K trigger paths is a catalyst cascade; the total classification power is $1 - \prod_i (1 - \kappa_i)$. Saturating the cascade ($\kappa \rightarrow 1$) requires $\sum_i \kappa_i = \infty$, which occurs when the trigger menu's composition depth reaches n_P .

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